

Phase 1: Programming Fundamentals (2-3 Weeks)

You should become comfortable with:

- Python
- Variables
- Functions
- OOP
- File Handling
- Exception Handling
- Modules
- Virtual Environments
- Pip

Libraries:

- NumPy
- Pandas
- Matplotlib

Projects:

- CSV Analyzer
 - Log Parser
 - Data Cleaner
 - Automation Scripts
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Phase 2: SQL & Databases

Learn:

- SELECT
- JOIN
- GROUP BY
- HAVING
- Window Functions
- CTE

- Indexes

Databases:

- PostgreSQL
- MySQL

Also learn:

- Redis
- Vector Databases

Examples:

- Pinecone
 - Qdrant
 - Chroma
 - Milvus
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Phase 3: Mathematics (Enough for AI)

You don't need a PhD.

Learn:

- Linear Algebra
- Matrix Multiplication
- Dot Product
- Vectors
- Probability
- Statistics
- Calculus Basics
- Gradient Descent

Understand:

- Why embeddings work
 - Similarity Search
 - Cosine Similarity
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Phase 4: Machine Learning

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- XGBoost
- SVM
- KNN
- Clustering
- PCA

Concepts:

- Overfitting
- Underfitting
- Bias
- Variance
- Feature Engineering
- Cross Validation

Libraries:

- scikit-learn

Projects:

- House Price Prediction
- Spam Detection
- Customer Churn
- Sales Forecast

Phase 5: Deep Learning

Learn:

- Neural Networks
- Activation Functions
- Backpropagation

- CNN
- RNN
- LSTM
- GRU
- Attention
- Transformers

Frameworks:

- PyTorch
- TensorFlow

Projects:

- Image Classification
 - Sentiment Analysis
 - OCR
 - Text Classification
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Phase 6: NLP

Understand:

- Tokenization
- Embeddings
- Word2Vec
- GloVe
- BERT
- Transformers
- Prompt Engineering

Libraries:

- Hugging Face Transformers
 - spaCy
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Phase 7: LLM Fundamentals

This is where most people start directly—and get confused.

Learn:

- Tokens
- Context Window
- Attention
- Positional Encoding
- Embeddings
- Decoder
- Sampling
- Temperature
- Top-K
- Top-P
- KV Cache
- Hallucination
- Quantization
- Fine Tuning
- LoRA
- RLHF
- Model Evaluation

Understand:

- How ChatGPT works
 - Why Claude is good at long context
 - Why Gemini can process multiple modalities
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Phase 8: Prompt Engineering

Learn:

- Zero Shot
 - One Shot
 - Few Shot
 - Chain of Thought
 - Tree of Thoughts
 - Self Consistency
 - ReAct
 - Prompt Templates
 - System Prompts
 - Output Formatting
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Phase 9: Retrieval-Augmented Generation (RAG)

Concepts:

- Chunking
- Embeddings
- Retrieval
- Hybrid Search
- Metadata
- Re-ranking
- Context Compression
- Citation
- Vector Search

Libraries:

- LangChain
- LlamaIndex

Projects:

- PDF Chatbot
 - Company Knowledge Assistant
 - HR Bot
 - Medical Search
 - Legal Assistant
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Phase 10: AI Agents

Concepts:

- Planning
- Memory
- Reflection
- Tool Calling
- Multi-step Reasoning
- Agent Loops
- Multi-Agent Systems

Frameworks:

- LangGraph
- AutoGen
- CrewAI

Projects:

- Research Agent
 - Coding Agent
 - QA Agent
 - Finance Agent
 - Customer Support Agent
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Phase 11: MCP (Model Context Protocol)

Learn:

- MCP Architecture
- MCP Servers
- MCP Clients
- Resources
- Prompts
- Tools
- Authentication
- Custom MCP Server Development

Projects:

- Jira MCP
 - GitHub MCP
 - Slack MCP
 - Gmail MCP
 - Database MCP
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Phase 12: AI Engineering

Learn:

- FastAPI
- REST APIs
- Docker
- Kubernetes
- Async Programming
- Streaming
- Authentication
- Caching
- Message Queues

Deploy:

- AI APIs
 - Agent Services
 - RAG Applications
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Phase 13: MLOps & LLMOps

Concepts:

- Model Versioning
- Experiment Tracking
- Monitoring
- Evaluation
- Prompt Versioning
- Cost Tracking
- Latency Monitoring
- A/B Testing

Tools:

- MLflow
 - Weights & Biases
 - LangSmith
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Phase 14: Cloud

Learn one cloud deeply:

- Amazon Web Services
- Microsoft Azure
- Google Cloud

Services:

- Compute
 - Storage
 - Networking
 - IAM
 - Serverless
 - AI Services
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Phase 15: Real Production Projects

Build at least 10 projects such as:

1. AI Resume Analyzer
 2. AI Interview Assistant
 3. AI QA Engineer
 4. AI Code Reviewer
 5. AI Email Assistant
 6. AI Sales Agent
 7. AI Medical Assistant
 8. AI Research Assistant
 9. AI Customer Support Agent
 10. AI Workflow Automation Platform
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Bonus Skills

- Git & GitHub
- Linux
- Docker
- CI/CD
- System Design
- Distributed Systems
- Design Patterns
- Testing

- Security
- Observability